

Applied Optimization

Exercise 6 - Newton's Method

Affine Invariance (1 + Bonus 0.5 pt)

(a) Show that Newton's method is invariant under affine transformations. Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a twice differentiable function. Consider the function $g(y) = f(Ay + b)$, where A is a non-singular constant matrix and $A \in \mathbb{R}^{n \times n}, b \in \mathbb{R}$. Prove that the sequence of points $\{x^i\}$ of f and the sequence of points $\{y^i\}$ of g generated by Newton's Method, starting from x^0 and y^0 respectively with $x^0 = Ay^0 + b$, have one-to-one correspondence under the transformation, i.e. $f(x^i) = g(y^i)$. Here we assume that the step lengths of the search directions of both functions are equal.

Solution:

Let $x = Ay + b$ be an affine transformation. To show that the Newton method is invariant to such transformations we first analyze the Newton step, and then the sequence.

The functions f, g and their derivatives are related by the following expressions

$$\begin{aligned}g(y) &= f(x) = f(Ay + b) \\ \nabla g(y) &= A^T \nabla f(Ay + b) \\ \nabla^2 g(y) &= A^T \nabla^2 f(Ay + b) A\end{aligned}$$

from the definition of the Newton steps it follows:

$$\begin{aligned}\Delta x_{nt} &= -(\nabla^2 f(x))^{-1} \nabla f(x) \\ \Delta y_{nt} &= -(\nabla^2 g(y))^{-1} \nabla g(y),\end{aligned}$$

Thus the Newton steps are related by

$$\begin{aligned}\Delta y_{nt} &= -(\nabla^2 g(y))^{-1} \nabla g(y) \\ &= -(A^T \nabla^2 f(Ay + b) A)^{-1} A^T \nabla f(Ay + b) \\ &= -A^{-1} \nabla^2 f(Ay + b)^{-1} (A^T)^{-1} A^T \nabla f(Ay + b) \\ &= A^{-1} \Delta x_{nt}\end{aligned}$$

The Newton steps are related by the derivative of the affine transformation $A \Delta y_{nt} = \Delta x_{nt}$, thus are equivalent.

For the sequences starting at x^0 and y^0 respectively, related by the affine transformation $x^0 = Ay^0 + b$ it holds, by the definition of the Newton update

$$\begin{aligned}x^1 &= x^0 + \Delta x^0 \\ &\vdots \\ x^{k+1} &= x^k + \Delta x^k\end{aligned}$$

where t for $i = 0 \dots k + 1$ is the step length at the i -th iteration. Using what we know about the transformation we can write the sequence for y :

$$\begin{aligned} y^1 &= y^0 + s^0 \Delta y^{(0)} \\ y^1 &= A^{-1}(x^0 - b) + s^0 A^{-1} \Delta x^0 \\ &\vdots \\ y^{k+1} &= A^{-1}(x^k - b) + s^k A^{-1} \Delta x^k \end{aligned}$$

Since the starting points are related by the affine transformation, the Newton updates are equivalent and the step length $t^i = s^i$, we have the one to one mapping of the two sequences.

$$\begin{aligned} y^{k+1} &= A^{-1}(x^k - b + t^k \Delta x^k) \\ &= A^{-1}(x^{k+1} - b). \end{aligned}$$

(b) Under affine transformations, show that the Newton's method with backtracking line search generates the same step length, i.e. the step length s^i of the function $f(x)$ equals to t^i of the function $g(y)$. In addition, show that the newton decrements of both functions are identical. (**Bonus (0.5 pt)**)

Solution:

In the backtracking line search, we need to find a t initialized as 1 that satisfies $f(x + t\Delta x) \leq f(x) + \alpha t \nabla f(x)^T \Delta x$ and $t > 0$, where α is a positive constant.

With the affine transformation $x = Ay + b$,

$$\begin{aligned} f(x + t\Delta x) &\leq f(x) + \alpha t \nabla f(x)^T \Delta x \\ \iff f(Ay + b + tA\Delta y) &\leq f(Ay + b) + \alpha t \nabla f(Ay + b)^T A\Delta y \\ \iff f(A(y + t\Delta y) + b) &\leq f(Ay + b) + \alpha t \nabla f(Ay + b)^T A\Delta y \\ \iff g(y + t\Delta y) &\leq g(y) + \alpha t \nabla g(y)^T (A^{-T})^T A\Delta y \\ \iff g(y + t\Delta y) &\leq g(y) + \alpha t \nabla g(y)^T \Delta y \end{aligned}$$

Hence, the step lengths of both functions are equal.

Finally, we check the stop criterion which is related to Newton decrement:

$$\begin{aligned} \lambda_g &= (\nabla g(y)^T \nabla^2 g(y)^{-1} \nabla g(y))^{1/2} \\ &= (\nabla f(Ay + b)^T A A^{-1} \nabla^2 f(Ay + b)^{-1} A^{-T} A^T \nabla f(Ay + b))^{1/2} \\ &= (\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x))^{1/2} \\ &= \lambda_f \end{aligned}$$